Campus Helpdesk

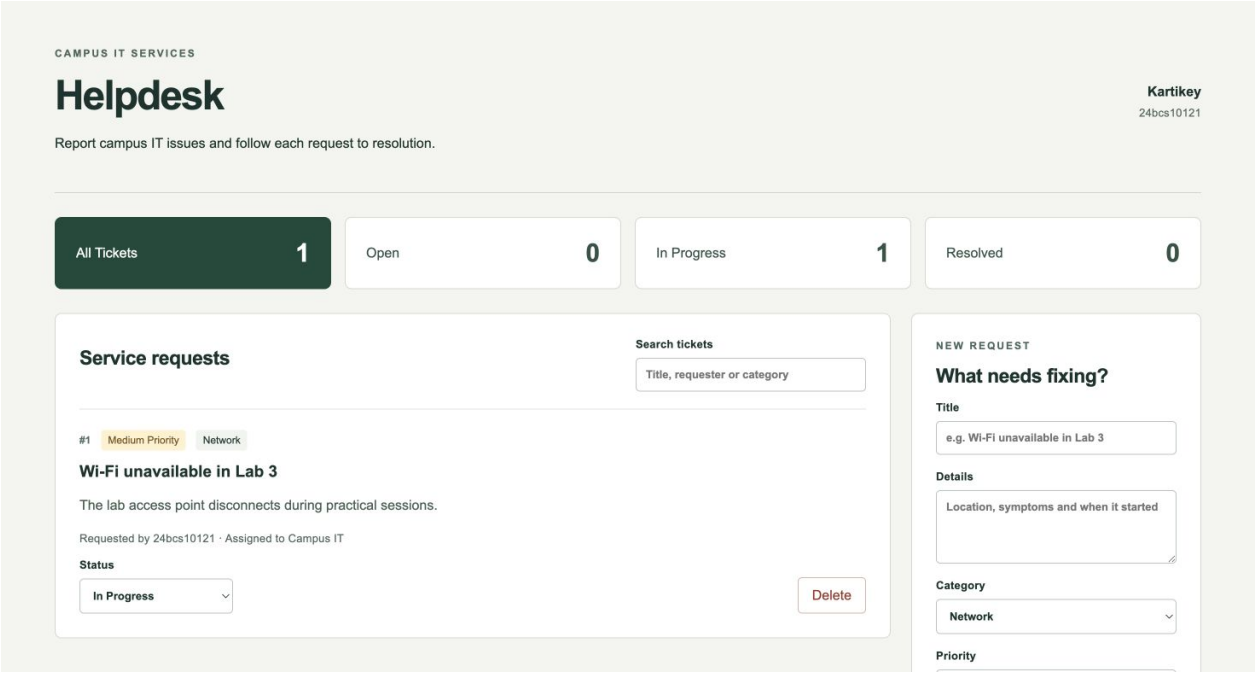
Campus IT requests, from report to resolution

Purpose

Track campus support requests with category, priority, assignee and status.

Submission

React, FastAPI and PostgreSQL with a tested DevSecOps delivery workflow.



Technology stack

Application

React + Vite, FastAPI, SQLAlchemy, PostgreSQL and Alembic

Delivery

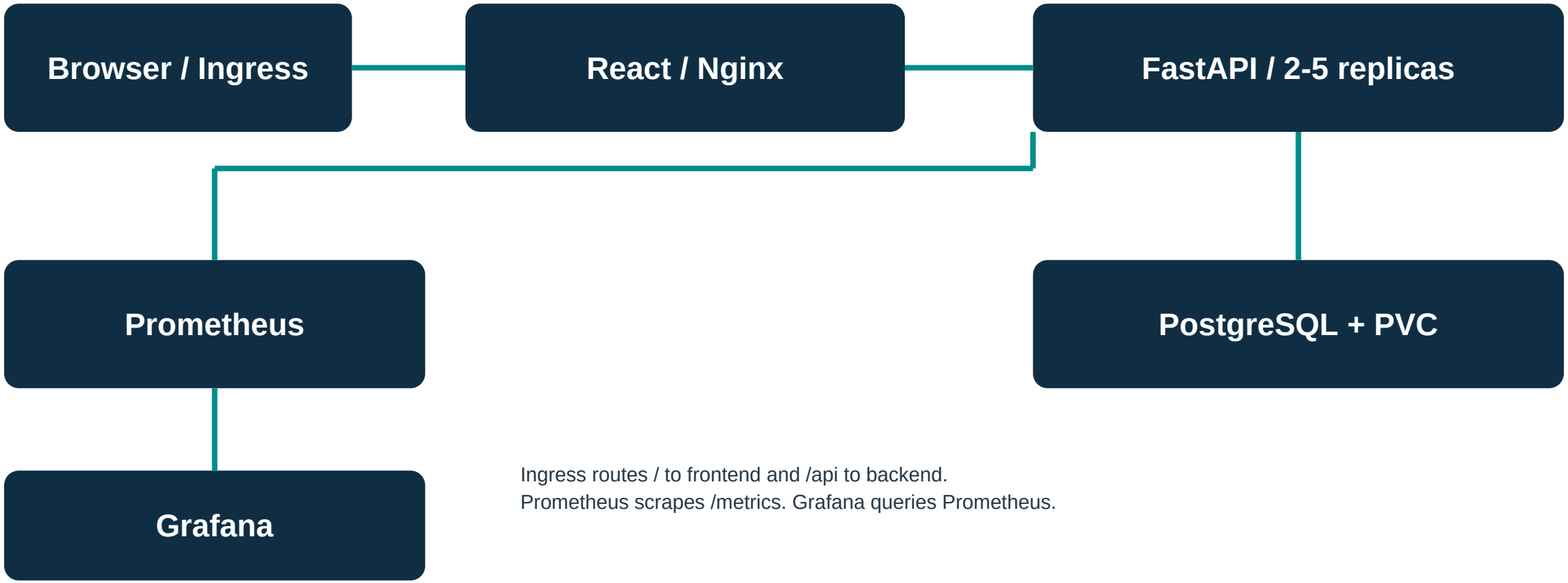
Pytest, Docker Compose, GitHub Actions, Trivy and GHCR

Infrastructure and operations

Terraform, AWS EKS, Kubernetes, Helm, Argo CD, Prometheus and Grafana

Source: session21-final-devops-project/README.md

Application architecture



Ingress routes / to frontend and /api to backend.
Prometheus scrapes /metrics. Grafana queries Prometheus.

Source: Helm templates, Compose configuration and monitoring values

API, migrations and tests

REST operations

Create, list, read, update and delete tickets. Stats group requests by status.

Database lifecycle

Alembic creates the tickets table. A migration job completes before API readiness.

17 isolated API tests

CRUD, validation, missing IDs, readiness and metrics. Each test uses a fresh SQLite database.

Source: application/backend/app, tests and alembic/versions

CI/CD flow

1. Quality gates

Pytest and frontend build, Bandit, dependency audits and Gitleaks

2. Build and publish

Build both images. Trivy blocks HIGH/CRITICAL findings. Publish GHCR tags using the Git SHA.

3. Deployment verification

Install with Helm in Kind. Check readiness, frontend HTTP and ticket creation. Export release values.

Container security

Small, non-root runtime images

Python Alpine backend and a multi-stage Node build with unprivileged Nginx runtime

A real release gate

An earlier Debian image failed its vulnerability scan. Changing the runtime removed affected packages.

Meaning of a clean scan

Both recorded release images had zero HIGH/CRITICAL findings at scan time. New advisories can change that result.

Source: security/README.md and evidence/main-ci-security-summary.txt

AWS infrastructure

Terraform

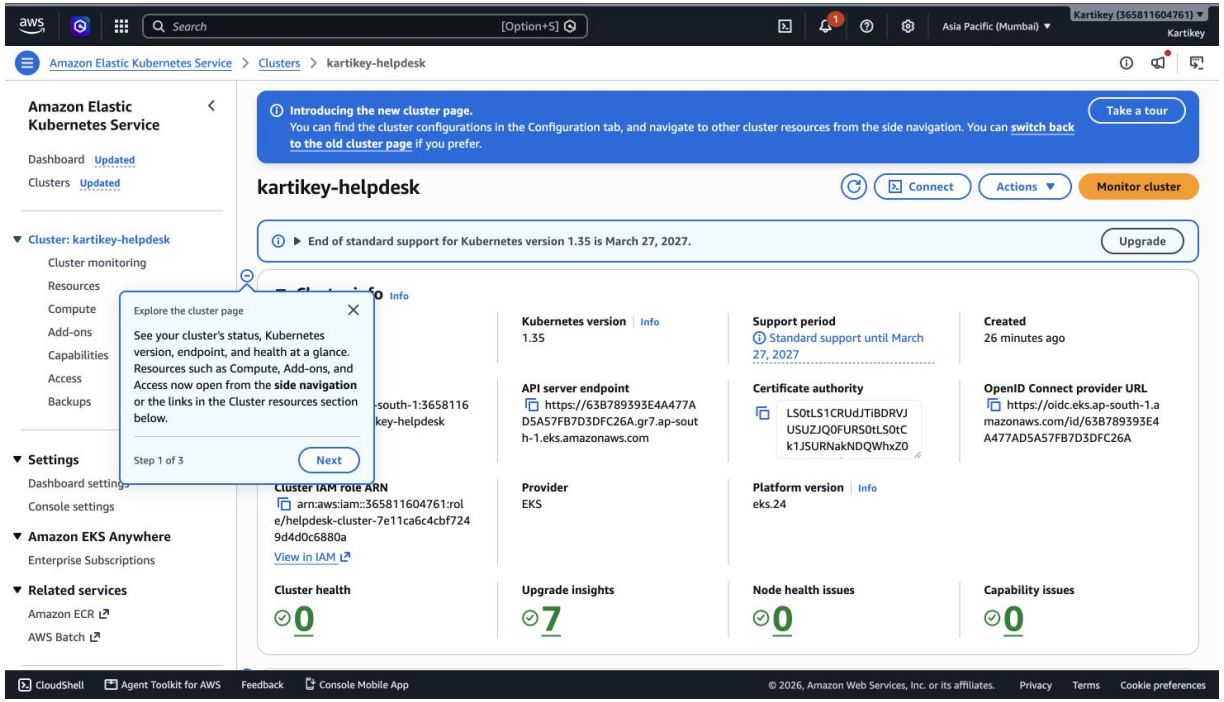
20 resources in ap-south-1. VPC, two public subnets and a managed EKS worker group.

Cluster access

Two t3.medium workers. Private worker API access and a restricted administrator CIDR.

Storage

EBS CSI uses Pod Identity to provision persistent PostgreSQL storage.



Kubernetes and Helm

Workloads

Two frontend and two backend replicas, plus a PostgreSQL StatefulSet and PVC.

Traffic and health

ClusterIP Services, Ingress routes, startup/liveness/readiness probes and CPU HPA.

Configuration

ConfigMap for settings. Runtime Secret for the database password. Helm packages the deployment.

Source: helm/helpdesk and evidence/eks-verification.txt | Recorded EKS run

GitOps release promotion

Desired state in Git

Argo CD reads Helm values, synchronizes changes and repairs drift.

Immutable release

Commit 9d7f9c4 promoted images built from 328f4c5. Argo reported Synced and Healthy.

Ownership

Helm bootstraps the application. Argo owns reconciliation after handoff. HPA owns backend replica count.

Source: gitops/README.md and evidence/eks-gitops-release.json | Recorded EKS run

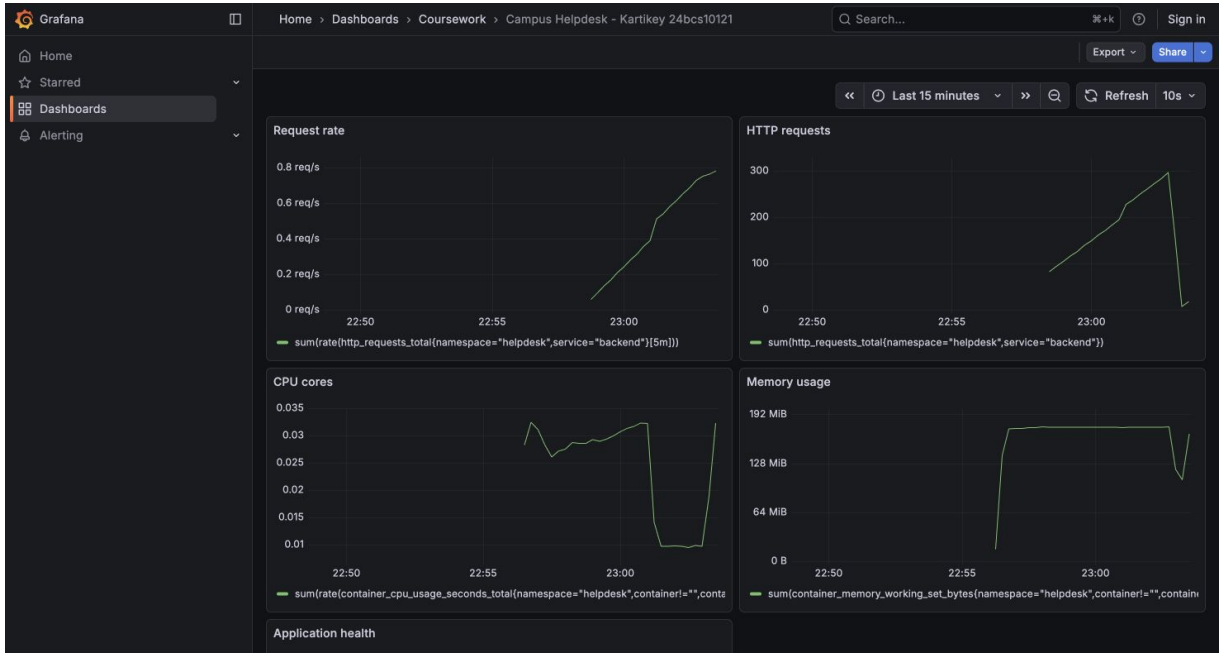
Prometheus and Grafana

Collection

ServiceMonitor discovers both backend Pods. Both targets reported UP.

Dashboards

Live request, CPU and memory panels. Metrics selectors match the Operator labels.



Source: monitoring/ and screenshots/eks-grafana.png | Recorded EKS run

Recovery and cleanup

Failure investigations

Image pull, Service selector and readiness faults. Capture diagnostics, restore configuration and verify HTTP.

Cloud teardown

The saved destroy plan removed all 20 project resources. Cleanup checks found no remaining project VPC, workers or volumes.

Evidence boundary

AWS screenshots show the 7 October run. The EKS cluster is now deleted. A new live cloud demo requires redeployment.

Source: troubleshooting/ and evidence/aws-destroy.txt, aws-cleanup.json

Key learnings and limits

Readiness protects traffic

A Running Pod may still be waiting for its database or migration.

GitOps changes operations

Pause self-healing during deliberate failure exercises and restore it afterward.

Classroom scope

No authentication or TLS. Monitoring storage is ephemeral. Production would need access control, backups and durable telemetry.